1) 1HP motor works 3 hours per day.

$$\text{Energy} = \frac{1\times746\times3}{1000} = \frac{2238}{1000}\text{Wh}$$
$$= 2.238 \text{ kwh}$$
$$= 2.238 \text{ unit per day}$$

2) 40w, 10 lamps glow 5 hours per day.

$$\text{Energy} = \frac{40\times10\times5}{1000} = \frac{2000}{1000}\text{Wh}$$
$$= 2 \text{ kwh}$$
$$= 2 \text{ units per day}$$

3) 900w heater works 2 hours per day.

$$\text{Energy} = \frac{900\times2}{1000} = \frac{1800}{1000}\text{Wh}$$
$$= 1.8 \text{ kwh}$$
$$= 1.8 \text{ units per day}$$

4) Total number of units consumed in 60 days = 60 (2.238 + 2 + 1.8)

$$= 60\left(2.238+2+1.8\right)$$
$$= 60\times6.038$$
$$= 362.28$$

Total number of unit = 362

Unit limit	Charge per unit	Usage unit	Cost in Rs
1–100	NIL	100	NIL
101–200	3.50	3.50 × 100	350
201–500	4.60	4.60 × 162	745.20
		Total	1095.20

Cost of electric charge for 60 days = Rs. 1095.20/-

Important note

Current - I – Ampere
Voltage – V – Volts
Resistance – R – Ω(ohms)
Emf – Volts
P.D – Volts
Power – P – watts
Ohms Law I = V/R
V = IR
R = V/I
Power $P = I^2R$
P = V × I

Series circuit I is common to all load, $V= V_1+V_2+V_3+\ldots.$, $R=R_1+R_2+R_3+\ldots\ldots$

Parallel circuit V is common to all load.

$$I = I_1 + I_2 + I_3\ldots.$$

$$\text{Resistance } R = \frac{R_1R_2R_3}{R_2R_3 + R_1R_3 + R_1R_2}$$

(if 3 loads)

Kirchoff's

i) Current Law $= I_1 + I_2 + I_3 = I_4 + I_5$ (or) $I_1 + I_2 + I_3 - I_4 - I_5 = 0$
ii) Voltage Law $V= IR_1 + IR_2 + IR_3$ ie $V= V_1 + V_2 + V_3$

Power Calculation

Power consumed = KW × hour = Kwh = 1unit.

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